Ps 06

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1 Problem 19 DONE

1.1 Task

1. (18) Linear systems of equations, Reduced row echelon form

(a) (12) Solve the following linear system of equations using Gaussian elimination, i.e., an LU-decomposition (possibly with partial pivoting)

$(2+i)z_1 + (-1+3i)z_2 + (4-i)z_3 + (1+2i)z_4 = 5+24i$, $(3-2i)z_1 + (2+i)z_2 + (-1+i)z_3 + 5z_4 = 24-7i$, $(-2+i)z_1 + (1-4i)z_2 + (3+2i)z_3 + (-1+i)z_4 = -14-2i$, $(1+3i)z_1 - 2z_2 + (2-i)z_3 + (4+i)z_4 = 7+10i$.

(b) (6) Transform the following real 4×5 matrix into reduced row echelon form:

$$A = \begin{pmatrix} 12 & 6 & 4 & -7 & -3 \\ -6 & -3 & 12 & 7 & 1 \\ 6 & 3 & -8 & -6 & 4 \\ 18 & 9 & 4 & -11 & 2 \end{pmatrix}$$

1.2 Solution

- Part (a)
  - Step 0: Given

$$A = \begin{bmatrix} 2+i & -1+3i & 4-i & 1+2i \\ 3-2i & 2+i & -1+i & 5 \\ -2+i & 1-4i & 3+2i & -1+i \\ 1+3i & -2 & 2-i & 4+i \end{bmatrix}, \quad b = \begin{bmatrix} 5+24i \\ 24-7i \\ -14-2i \\ 7+10i \end{bmatrix}$$

  - Step 1: Pivot
    We examine the first column to find the pivot element with the largest absolute value.

$$\begin{aligned} |a_{11}| &= |2+i| = \sqrt{5} \\ |a_{21}| &= |3-2i| = \sqrt{13} \\ |a_{31}| &= |-2+i| = \sqrt{5} \\ |a_{41}| &= |1+3i| = \sqrt{10} \end{aligned}$$

    The maximum absolute value is in the second row ($\sqrt{13}$). We swap row 1 and row 2 ($R_1 \leftrightarrow R_2$).

$$
\begin{aligned}
P_1 A &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A \\
&= \begin{bmatrix} 3-2i & 2+i & -1+i & 5 \\ 2+i & -1+3i & 4-i & 1+2i \\ -2+i & 1-4i & 3+2i & -1+i \\ 1+3i & -2 & 2-i & 4+i \end{bmatrix}
\end{aligned}
$$

- Step 2: Eliminate column 1
  We compute the multipliers for the first column ($l_{i1} = a_{i1}/a_{11}$). For example:
  $$l_{21} = \frac{2+i}{3-2i} = \frac{(2+i)(3+2i)}{13} = \frac{4+7i}{13}$$
  We perform the row operations $R_i \leftarrow R_i - l_{i1}R_1$ for $i = 2, 3, 4$. This leaves zeros in the first column below the diagonal.
- Step 3: Iterate
  We repeat the pivoting and elimination process for the submatrices (Columns 2 and 3).
  1. Find $\max |a_{i2}^{(2)}|$ for $i \geq 2$, swap rows if necessary.
  2. Eliminate column 2.
  3. Find $\max |a_{i3}^{(3)}|$ for $i \geq 3$, swap rows if necessary.
  4. Eliminate column 3.

  This yields the factorization $PA = LU$.
- Step 4: Solve
  We solve the system in two stages using the computed factors:
  1. **Forward Substitution:** Solve $Ly = Pb$ for y.
  2. **Backward Substitution:** Solve $Ux = y$ for x.

  Performing these substitutions yields the solution vector $x \in \mathbb{C}^4$.

- Part (b)
  - Step 0: Given
  $$A = \begin{bmatrix} 12 & 6 & 4 & -7 & -3 \\ -6 & -3 & 12 & 7 & 1 \\ 6 & 3 & -8 & -6 & 4 \\ 18 & 9 & 4 & -11 & 2 \end{bmatrix}$$
  - Step 1: Pivot Col 1
    Swap R_1 with R_4 (Pivot $18 > 12$).
  $$A \xrightarrow{R_1 \leftrightarrow R_4} \begin{bmatrix} 18 & 9 & 4 & -11 & 2 \\ -6 & -3 & 12 & 7 & 1 \\ 6 & 3 & -8 & -6 & 4 \\ 12 & 6 & 4 & -7 & -3 \end{bmatrix}$$

- Step 2: Eliminate Col 1

$$\xrightarrow[R_4 \leftarrow R_4 - \frac{2}{3}R_1]{\substack{R_2 \leftarrow R_2 + \frac{1}{3}R_1 \\ R_3 \leftarrow R_3 - \frac{1}{3}R_1}} \begin{bmatrix} 18 & 9 & 4 & -11 & 2 \\ 0 & 0 & \frac{40}{3} & \frac{10}{3} & \frac{5}{3} \\ 0 & 0 & -\frac{28}{3} & -\frac{7}{3} & \frac{10}{3} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & -\frac{13}{3} \end{bmatrix}$$

Scale rows to simplify (integers preferred for inspection): $R_2 \leftarrow \frac{3}{5}R_2 \implies [0, 0, 8, 2, 1]$ $R_3 \leftarrow 3R_3 \implies [0, 0, -28, -7, 10]$ $R_4 \leftarrow 3R_4 \implies [0, 0, 4, 1, -13]$

- Step 3: Pivot Col 3

  The second column contains only zeros below the pivot, so we advance to Column 3. Pivot candidates: $8, -28, 4$. The maximum absolute value is $|-28|$. Swap R_2 and R_3.

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 18 & 9 & 4 & -11 & 2 \\ 0 & 0 & -28 & -7 & 10 \\ 0 & 0 & 8 & 2 & 1 \\ 0 & 0 & 4 & 1 & -13 \end{bmatrix}$$

- Step 4: Eliminate Col 3

  Note that in R_3: $8 = -\frac{2}{7}(-28)$. Note that in R_4: $4 = -\frac{1}{7}(-28)$.

$$\xrightarrow[R_4 \leftarrow R_4 + \frac{1}{7}R_2]{R_3 \leftarrow R_3 + \frac{2}{7}R_2} \begin{bmatrix} 18 & 9 & 4 & -11 & 2 \\ 0 & 0 & -28 & -7 & 10 \\ 0 & 0 & 0 & 0 & \frac{27}{7} \\ 0 & 0 & 0 & 0 & -\frac{81}{7} \end{bmatrix}$$

- Step 5: Pivot Col 5

  We inspect the next available column (Col 5) for the remaining rows. We eliminate R_4 using R_3. Note that $-\frac{81}{7} = -3\left(\frac{27}{7}\right)$.

$$\xrightarrow{R_4 \leftarrow R_4 + 3R_3} \begin{bmatrix} 18 & 9 & 4 & -11 & 2 \\ 0 & 0 & -28 & -7 & 10 \\ 0 & 0 & 0 & 0 & \frac{27}{7} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- Step 6: Normalize to RREF
  1. **Scale pivots to 1:** $R_3 \leftarrow \frac{7}{27}R_3 \implies [0, 0, 0, 0, 1]$ $R_2 \leftarrow -\frac{1}{28}R_2 \implies [0, 0, 1, \frac{1}{4}, -\frac{5}{14}]$ $R_1 \leftarrow \frac{1}{18}R_1 \implies [1, \frac{1}{2}, \frac{2}{9}, -\frac{11}{18}, \frac{1}{9}]$
  2. **Backward Elimination (Eliminate entries above pivots):**
     * Use R_3 (Col 5) to eliminate entries in R_1, R_2. $R_2 \leftarrow R_2 + \frac{5}{14}R_3 \implies [0, 0, 1, \frac{1}{4}, 0]$ $R_1 \leftarrow R_1 - \frac{1}{9}R_3 \implies [1, \frac{1}{2}, \frac{2}{9}, -\frac{11}{18}, 0]$
     * Use R_2 (Col 3) to eliminate entry in R_1. $R_1 \leftarrow R_1 - \frac{2}{9}R_2$ Entry (1,4): $-\frac{11}{18} - \frac{2}{9}(\frac{1}{4}) = -\frac{11}{18} - \frac{1}{18} = -\frac{12}{18} = -\frac{2}{3}$.

Final Result

$$\text{RREF}(A) = \begin{bmatrix} 1 & \frac{1}{2} & 0 & -\frac{2}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2 Problem 20 DONE

2.1 Task

1. (15) Vector space

(a) (9) Let V be an $\mathbb{F}$ -vector space and X an arbitrary set. Consider the set $\mathcal{F}(X, V)$ of all functions $f : X \to V$ with pointwise operations as defined in the lecture course. Show that $\mathcal{F}(X, V)$ is a $\mathbb{F}$ -vector space.

(b) (6) Let V and W be $\mathbb{F}$ -vector spaces. Consider on the cartesian product $V \times W$ the following operations: For all $v, v' \in V, w, w' \in W$ and $\lambda \in \mathbb{F}$ let

$$(v, w) + (v', w*) := (v + v', w + w')$$
$$\lambda(v, w) := (\lambda v, \lambda w)$$

Show that these two operations make $V \times W$ a vector space over $\mathbb{F}$.

2.2 Solution

- Part (a)
  - Step 0: Definitions
    Let $\mathcal{F}(X, V) = \{f \mid f : X \to V\}$. The operations are defined pointwise for all $x \in X$:
    1. Addition: $(f + g)(x) := f(x) +_V g(x)$
    2. Scalar Multiplication: $(\lambda f)(x) := \lambda \cdot_V f(x)$

    To prove $\mathcal{F}(X, V)$ is a vector space, I must verify the 8 vector space axioms. Since V is a vector space, its operations $+_V$ and $\cdot_V$ satisfy these axioms.
  - Step 1: Abelian Group Properties (Addition)
    * **(i) Associativity:** For all $f, g, h \in \mathcal{F}(X, V)$ and $x \in X$:

$$\begin{aligned} ((f + g) + h)(x) &= (f + g)(x) +_V h(x) \\ &= (f(x) +_V g(x)) +_V h(x) \\ &= f(x) +_V (g(x) +_V h(x)) \qquad \text{(Associativity in } V) \\ &= f(x) +_V (g + h)(x) \\ &= (f + (g + h))(x) \end{aligned}$$

Thus, $(f+g)+h = f+(g+h)$.

* **(ii) Commutativity:**

$$\begin{aligned}(f+g)(x) &= f(x) +_V g(x) \\ &= g(x) +_V f(x) \qquad \text{(Commutativity in } V\text{)} \\ &= (g+f)(x)\end{aligned}$$

* **(iii) Identity Element:** Let $\mathbf{0} \in \mathcal{F}(X,V)$ be the zero function defined by $\mathbf{0}(x) = 0_V$ for all $x \in X$.

$$\begin{aligned}(f+\mathbf{0})(x) &= f(x) +_V \mathbf{0}(x) \\ &= f(x) +_V 0_V \\ &= f(x) \qquad \text{(Identity in } V\text{)}\end{aligned}$$

  Thus, $\mathbf{0}$ is the additive identity.

* **(iv) Inverse Element:** For any f, define $(-f)$ by $(-f)(x) = -f(x)$ (the additive inverse in V).

$$\begin{aligned}(f+(-f))(x) &= f(x) +_V (-f)(x) \\ &= 0_V \qquad \text{(Inverse in } V\text{)} \\ &= \mathbf{0}(x)\end{aligned}$$

- Step 2: Scalar Multiplication Properties
  Let $\lambda, \mu \in \mathbb{F}$ and $f, g \in \mathcal{F}(X,V)$.

* **(v) Distributivity (Scalar sum):**

$$\begin{aligned}((\lambda+\mu)f)(x) &= (\lambda+\mu) \cdot_V f(x) \\ &= \lambda \cdot_V f(x) +_V \mu \cdot_V f(x) \qquad \text{(Distributivity in } V\text{)} \\ &= (\lambda f)(x) +_V (\mu f)(x) \\ &= (\lambda f + \mu f)(x)\end{aligned}$$

* **(vi) Distributivity (Vector sum):**

$$\begin{aligned}(\lambda(f+g))(x) &= \lambda \cdot_V (f(x) +_V g(x)) \\ &= \lambda \cdot_V f(x) +_V \lambda \cdot_V g(x) \qquad \text{(Distributivity in } V\text{)} \\ &= (\lambda f + \lambda g)(x)\end{aligned}$$

* **(vii) Associativity of scalar multiplication:**

$$\begin{aligned}((\lambda\mu)f)(x) &= (\lambda\mu) \cdot_V f(x) \\ &= \lambda \cdot_V (\mu \cdot_V f(x)) \qquad \text{(Associativity in } V\text{)} \\ &= \lambda \cdot_V (\mu f)(x) \\ &= (\lambda(\mu f))(x)\end{aligned}$$

* **(viii) Unitary Element:**

$$\begin{aligned}(1 \cdot f)(x) &= 1 \cdot_V f(x) \\ &= f(x) \qquad \text{(Identity in } V\text{)}\end{aligned}$$

- Step 3: Conclusion
  Since all 8 axioms hold pointwise due to the vector space structure of V, $\mathcal{F}(X, V)$ is a vector space over $\mathbb{F}$.

- Part (b)
  - Step 0: Given
    We check the axioms for $U = V \times W$ with elements $u = (v, w)$. Addition: $(v, w) + (v', w') = (v + v', w + w')$ Scalar Mult: $\lambda(v, w) = (\lambda v, \lambda w)$
  - Step 1: Additive Properties
    * **(i) Associativity & (ii) Commutativity:** These follow immediately from the component-wise associativity and commutativity in V and W. Example (Commutativity):
    $$(v, w) + (v', w') = (v + v', w + w') = (v' + v, w' + w) = (v', w') + (v, w)$$
    * **(iii) Identity:** Let $0_U = (0_V, 0_W)$.
    $$(v, w) + (0_V, 0_W) = (v + 0_V, w + 0_W) = (v, w)$$
    * **(iv) Inverse:** Let $-u = (-v, -w)$.
    $$(v, w) + (-v, -w) = (v + (-v), w + (-w)) = (0_V, 0_W) = 0_U$$
  - Step 2: Scalar Properties
    * **(v) Distributivity over scalar sum:**
    $$\begin{aligned}(\lambda + \mu)(v, w) &= ((\lambda + \mu)v, (\lambda + \mu)w) \\ &= (\lambda v + \mu v, \lambda w + \mu w) \qquad \text{(Distributivity in } V, W) \\ &= (\lambda v, \lambda w) + (\mu v, \mu w) \\ &= \lambda(v, w) + \mu(v, w)\end{aligned}$$
    * **(vi) Distributivity over vector sum:**
    $$\begin{aligned}\lambda((v, w) + (v', w')) &= \lambda(v + v', w + w') \\ &= (\lambda(v + v'), \lambda(w + w')) \\ &= (\lambda v + \lambda v', \lambda w + \lambda w') \\ &= (\lambda v, \lambda w) + (\lambda v', \lambda w') \\ &= \lambda(v, w) + \lambda(v', w')\end{aligned}$$
    * **(vii) Scalar associativity:**
    $$\begin{aligned}(\lambda\mu)(v, w) &= ((\lambda\mu)v, (\lambda\mu)w) \\ &= (\lambda(\mu v), \lambda(\mu w)) \\ &= \lambda(\mu v, \mu w) \\ &= \lambda(\mu(v, w))\end{aligned}$$
    * **(viii) Unitary:**
    $$1(v, w) = (1v, 1w) = (v, w)$$
  - Step 3: Conclusion
    The set $V \times W$ equipped with component-wise operations satisfies all axioms and is therefore a vector space over $\mathbb{F}$.

3 Problem 21 DONE

3.1 Task

1. (12) Subspace

We consider the vector space $\mathbb{F}^n$ for the specified field and some $4 \leq n \in \mathbb{N}$. Check, whether the following subsets are subspaces.

(a) (2) $W_1 := \{x \in \mathbb{R}^n | x_1 + x_2 = 0\}$,

(b) (2) $W_2 := \{x \in \mathbb{R}^n | \sum_{j=1}^n jx_j = 0\}$,

(c) (2) $W_3 := \{x \in \mathbb{R}^n | x_1 - x_3 = x_4 + 1\}$,

(d) (2) $W_4 := \{x \in \mathbb{R}^n | \sum_{i=1}^n x_i^2 = 0\}$,

(e) (2) R$W_5 := \{x \in \mathbb{F}_2^3 | x_1^2 + x_2 + x_3^3 = 0\}$,

(f) (2) $W_6 := \{x \in \mathbb{C}^2 | x_1 \neq 0 \wedge \frac{x_2}{x_1} = 1\}$.

3.2 Solution

To prove a subset W is a subspace, I must check three conditions:

1. $0 \in W$.
2. Closure under addition ($u, v \in W \implies u + v \in W$).
3. Closure under scalar multiplication ($\lambda \in \mathbb{F}, u \in W \implies \lambda u \in W$).
4. Part (a)

   $W_1 = \{x \in \mathbb{R}^n \mid x_1 + x_2 = 0\}$

   - **Step 1: Check Zero Vector** For $x = 0$, $x_1 + x_2 = 0 + 0 = 0$. Thus, $0 \in W_1$.
   - **Step 2: Closure properties** This is the solution set to a homogeneous linear equation ($Ax = 0$ where row 1 is $[1, 1, 0, \ldots]$). Let $x, y \in W_1$ and $\lambda \in \mathbb{R}$.

   $$(x_1 + y_1) + (x_2 + y_2) = (x_1 + x_2) + (y_1 + y_2) = 0 + 0 = 0 \implies x + y \in W_1$$

   $$(\lambda x_1) + (\lambda x_2) = \lambda(x_1 + x_2) = \lambda(0) = 0 \implies \lambda x \in W_1$$

   - **Conclusion W_1 is a subspace**.
5. Part (b)

   $W_2 = \{x \in \mathbb{R}^n \mid \sum_{j=1}^n jx_j = 0\}$

   - **Step 1: Check Zero Vector** $\sum j(0) = 0$. Thus $0 \in W_2$.

- **Step 2: Linearity** This condition defines the kernel of the linear functional $f(x) = \sum jx_j$. The kernel of a linear map is always a subspace. Alternatively:

$$\sum j(x_j + y_j) = \sum jx_j + \sum jy_j = 0 + 0 = 0$$

$$\sum j(\lambda x_j) = \lambda \sum jx_j = 0$$

- **Conclusion W_2 is a subspace**.

6. Part (c)

   $W_3 = \{x \in \mathbb{R}^n \mid x_1 - x_3 = x_4 + 1\}$

   - **Step 1: Check Zero Vector** For the zero vector $x = 0$, plug in components:

   $$0 - 0 = 0 \quad \text{vs} \quad 0 + 1 = 1$$

   $$0 \neq 1$$

   The zero vector is not in W_3.

   - **Conclusion W_3 is NOT a subspace**.

7. Part (d)

   $W_4 = \{x \in \mathbb{R}^n \mid \sum_{i=1}^n x_i^2 = 0\}$

   - **Step 1: Analysis of constraint in $\mathbb{R}$** Since $x_i \in \mathbb{R}$, $x_i^2 \geq 0$. The sum of non-negative terms is zero if and only if every term is zero.

   $$\sum_{i=1}^n x_i^2 = 0 \iff x_i = 0 \quad \forall i$$

   Therefore, $W_4 = \{0\}$ (the set containing only the zero vector).

   - **Step 2: Trivial Subspace** The set $\{0\}$ is the trivial subspace. It satisfies all axioms $(0 + 0 = 0, \lambda 0 = 0)$.
   - **Conclusion W_4 is a subspace**.

8. Part (e)

   $W_5 = \{x \in \mathbb{F}_2^3 \mid x_1^2 + x_2 + x_3^3 = 0\}$

   - **Step 1: Field Properties of $\mathbb{F}_2$** In the field $\mathbb{F}_2 = \{0, 1\}$, we have the property that $a^k = a$ for all $a \in \mathbb{F}_2, k \geq 1$.

   $$0^2 = 0, 1^2 = 1 \implies x_1^2 = x_1$$

   $$0^3 = 0, 1^3 = 1 \implies x_3^3 = x_3$$

   The condition simplifies to:

   $$x_1 + x_2 + x_3 = 0$$

   - **Step 2: Linearity** This is a linear homogeneous equation over $\mathbb{F}_2$. As seen in parts (a) and (b), solution sets to linear homogeneous equations are subspaces.

- **Conclusion W_5 is a subspace**.

9. Part (f)

   $W_6 = \{x \in \mathbb{C}^2 \mid x_1 \neq 0 \wedge \frac{x_2}{x_1} = 1\}$

   - **Step 1: Check Zero Vector** The definition explicitly requires $x_1 \neq 0$. The zero vector $0 = (0, 0)$ has first component 0. Thus, $0 \notin W_6$.
   - **Conclusion W_6 is NOT a subspace**.

4 Problem 22 DONE

4.1 Task

1. (12) Spanning sets, Linear Independence

(a) (2) Show that $\mathbb{C}$ is a vector space over $\mathbb{R}$ and find a small spanning set for $\mathbb{C}$.

(b) (4) We consider the vector space $\mathbb{F}^4$ for some field $\mathbb{F}$. Show that the vectors $(1, 0, 0, 0)^T$, $(1, 1, 0, 0)^T$, $(1, 1, 1, 0)^T$ and $(1, 1, 1, 1)^T$ are linearly independent.

(c) (6) Show that the three vectors $(i, 1, 2+i)$, $(1, 3i+1, 2)$ and $(-1, 5i, 5)$ are a spanning set of $\mathbb{C}^3$. Are they linearly independent?

4.2 Solution

- Part (a)
  - Step 1: Vector Space Axioms We view $\mathbb{C}$ as the set of numbers $z = x + iy$. Since $\mathbb{C}$ is a field itself containing $\mathbb{R}$, the addition of complex numbers is associative, commutative, has an identity (0), and inverses. Scalar multiplication by $\lambda \in \mathbb{R}$ distributes over addition: $\lambda(z_1 + z_2) = \lambda z_1 + \lambda z_2$ and $(\lambda + \mu)z = \lambda z + \mu z$. Thus, $\mathbb{C}$ satisfies all axioms of a vector space over $\mathbb{R}$.
  - Step 2: Spanning Set Any element $z \in \mathbb{C}$ can be written uniquely as:

    $$z = a \cdot 1 + b \cdot i \quad \text{where } a, b \in \mathbb{R}$$

    Therefore, the set $\{1, i\}$ spans $\mathbb{C}$ over $\mathbb{R}$.

- Part (b)

  Let the given vectors be columns of a matrix A.

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, v_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

To show linear independence, we consider the equation:

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = \mathbf{0}$$

Writing this as a linear system $Ac = 0$:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

This is an upper triangular matrix with non-zero entries on the diagonal (all 1s).

- From Row 4: $1 \cdot c_4 = 0 \implies c_4 = 0$.
- From Row 3: $1 \cdot c_3 + 1 \cdot c_4 = 0 \implies c_3 = 0$.
- From Row 2: $1 \cdot c_2 + c_3 + c_4 = 0 \implies c_2 = 0$.
- From Row 1: $1 \cdot c_1 + c_2 + c_3 + c_4 = 0 \implies c_1 = 0$.

Since the only solution is the trivial solution $c = 0$, the vectors are **linearly independent**.

- Part (c)

  Let v_1, v_2, v_3 be the columns of the matrix A.

$$A = \begin{bmatrix} i & 1 & -1 \\ 1 & 3i+1 & 5i \\ 2+i & 2 & 5 \end{bmatrix}$$

  - Step 1: Gaussian Elimination to determine Rank We want to determine if the matrix has full rank (Rank = 3).

    Swap $R_1 \leftrightarrow R_2$ to get a pivot of 1:

$$\begin{bmatrix} 1 & 3i+1 & 5i \\ i & 1 & -1 \\ 2+i & 2 & 5 \end{bmatrix}$$

    Eliminate Column 1: $R_2 \leftarrow R_2 - iR_1$: $(2,2) : 1 - i(3i+1) = 1 - (-3+i) = 4 - i$ $(2,3) : -1 - i(5i) = -1 + 5 = 4$

    $R_3 \leftarrow R_3 - (2+i)R_1$: $(3,2) : 2 - (2+i)(3i+1) = 2 - (6i + 2 + 3i^2 + i) = 2 - (-1 + 7i) = 3 - 7i$ $(3,3) : 5 - (2+i)(5i) = 5 - (10i - 5) = 10 - 10i$

$$A' = \begin{bmatrix} 1 & 3i+1 & 5i \\ 0 & 4-i & 4 \\ 0 & 3-7i & 10-10i \end{bmatrix}$$

    Check for linear dependence between R_2 and R_3: If they were dependent, $(4-i)(10-10i) = 4(3-7i)$. LHS: $40 - 40i - 10i + 10i^2 = 30 - 50i$ RHS: $12 - 28i$ Since $30 - 50i \neq 12 - 28i$, the rows are linearly independent.

  - Step 2: Conclusion The matrix A has 3 pivots, so $\text{Rank}(A) = 3$. Since the dimension of $\mathbb{C}^3$ is 3 and we have 3 linearly independent vectors, they form a **basis**.

    1. Since they form a basis, they span $\mathbb{C}^3$.
    2. Yes, they are linearly independent.